

Vector clock-based race detection

A thread's **logical clock**: a non-negative integer. A thread's logical clock value increases each time the thread releases a mutex. Releasing a mutex indicates transitioning to next phase of concurrent program.

A **vector clock** is a tuple of N logical clocks, one per thread:

$(c_0, c_1, \dots, c_{N-1})$. Equivalently: vector clock is a mapping from Threads to natural numbers. If V is a vector clock, $V(t)$ denotes logical clock of thread t . Set of all vector clocks: VC